

Shamsher Tiwari

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EXPERIENCE

Dell Technologies (System Development Engineer 2)

Aug 2023-Present

- **Scope-Based Access Control & API Security:**
 - Designed and engineered a high-security backend API using **NodeJS, Express, and MongoDB**, implementing Scope-Based Access Control (SBAC) to provide granular permissions using Claude .
 - Leveraged **Claude AI** and large language models to architect complex permission matrices, which increased custom user control by **70%** and provided a critical security layer for managing AI-driven 17G servers via OpenManage Enterprise (OME) across global datacenters.
- **17G Backend Development & OME Integration:**
 - Collaborated with the development Team of the 17G backend API and its seamless integration with OpenManage Enterprise (OME), directly **impacting over 30% of datacentre administrators worldwide**.
 - Successfully integrated the 17G Virtual Console, enhancing remote management capabilities and system accessibility for enterprise-level hardware with 6+ AI tools.
- **Real-time Data Visualization (Kafka Connectivity):**
 - Developed an end-to-end frontend service **using ReactJS and Redux Toolkit** to manage Kafka Connectivity, enabling customers to stream and visualize large datasets in real time.
 - Implemented efficient data handling and utilized **AI code optimization techniques** to render high-volume telemetry, significantly improving data-driven decision-making and monitoring for cloud-scale infrastructure.
- **Automation of OME Deployment:**
 - Took complete ownership of an end-to-end Python automation framework built from scratch using **python and Dell Common Framework Libraries** to automate OME configurations.
 - Integrated **AI models** to generate and refine complex automation scripts, enabling a **2x increase in deployment speed** and reducing manual engineering effort by **50%** by automating complex backup and restoration workflows.

Draup (Front End Developer Intern)

Feb 2023 – May 2023

- Migrated email templates from backend to **Postmark**, reducing frontend dependency on backend for template edits.
- Utilized **JavaScript libraries** to streamline template management, improving efficiency and speeding up response to template changes.

PROJECTS

Shopee - Ecommerce websites

ReactJS, NodeJS, ExpressJS MongoDB, Redux, Stripe - [GitHub Link](#)

- Architected a scalable end-to-end shopping solution featuring secure **JWT-based authentication** and a **robust administrative** dashboard for order management.
- **Integrated Stripe API** for secure payment processing and **Cloudinary** for high-performance image hosting and optimization.
- Enhanced user experience by developing advanced server-side features including **pagination, fuzzy search**, and multi-parameter product **filtering**.

Chatly - Real-time Chat App

ReactJS, NodeJS, ExpressJS, MongoDB, Redux, Socket.IO - [GitHub Link](#)

- Built a **responsive, event-driven chat application** featuring **live messaging and real-time** active user status tracking **via Socket.IO**.
- **Enabled users to send** text, images, and emojis, **with file uploads managed via** Multer and **image storage through** Cloudinary.
- **Managed complex application state** using Redux to ensure seamless data synchronization across the UI during high-frequency interactions.

VocaLink - AI Voice enabled Virtual Agent

ReactJS, NodeJS, ExpressJS, MongoDB, Redux- [GitHub Link](#)

- Engineered a speech-recognition **AI chatbot** using the **MERN stack**, supporting **Indian languages** via **Web Speech API**, with real-time interaction and **intent detection** powered by **Gemini Flash 2.5**.
- Engineered a **JSON-based response** system to handle **Google/YouTube search/play, date/time/day/month, weather** and **calculator** based on natural language input.
- Enabled assistant customization (name/creator), stored **user inputs and assistant images** using **MongoDB** and **Cloudinary** and ensured smooth conversation flow with **Socket.IO** and multilingual support.

TECHNICAL SKILLS

- **Programming Languages / Skills-** HTML5, CSS, C/C++, Python, JavaScript (ES6+), SQL, NoSQL
- **Frameworks & Libraries-** ReactJS, Redux, ExpressJS, NodeJS, Tailwind CSS, Mongoose, Socket.io
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Cloud/ Tools** –GitHub, Copilot, Claude, ChatGPT, JIRA, vSphere Client, VMware ESXI, Microsoft System Center Consoles (SCVMM, MCM, SCOM), Hyper-V Manager, Cloudinary, Stripe Payment Gateway
- **Subjects:** Data structures and algorithms, Computer Networks, Object oriented Programming (OOPS), Software Development Lifecycle (SDLC), Agile methodology, operating system, Database Management System (DBMS), System Design, Server Architecture

EDUCATION

Dayananda Sagar College of Engineering

B.E. in Computer Science Engineering – 9.5 CGPA

Bengaluru, Karnataka

Aug 2019 – July 2023